

Muhammad Ammar

Machine Learning Engineer

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PROFILE

Machine Learning Engineer focused on real-time computer vision and deep learning deployment, with hands-on experience optimizing high-performance inference pipelines using PyTorch, TensorRT, and FastAPI. Delivers end-to-end ML systems — from data and modeling to deployed APIs on production-grade datasets.

EDUCATION

Mansoura University | Faculty of Computer and Information Science

2022 – 2026

Bachelor's Degree in Computer Science

EXPERIENCE

- **Machine Learning Intern**, Cellula Technology Oct 2025 – Dec 2025 (Project-based)
 - Categorized cancellation cases for hotel bookings across **500K+** entries, building a pipeline that achieved **87% accuracy and 0.91 F1-score** on the minority class.
 - Estimated Uber trip fares on **1M+** records, **achieving R^2 of 0.74 and MAE of 1.01** using gradient boosting.
 - Delivered both projects end-to-end — data cleaning, EDA, feature engineering, and modeling — then deployed each as a FastAPI REST API with a connected frontend for live predictions.

PROJECTS

- [Real-Time Industrial Quality Control System](#): **University Graduation Project**
Stack: YOLO, OpenCV, FastAPI, WebRTC, Firebase, ONNX, TensorRT
 - Designed a two-stage YOLO pipeline for automated defect detection, achieving **84.5% mAP50** on defect classification and **99.5% mAP50** on part localization.
 - Optimized the full pipeline from **30 FPS** to **170–200 FPS** by converting both models from PyTorch to TensorRT — a **~6x** throughput improvement.
 - Deployed a real-time monitoring dashboard with WebRTC live streaming, session management, and inspection insights aggregation.
 - Validated the system on a simulated production setup with a physical camera and miniature conveyor, demonstrating stable real-time performance.
- [Sign Language Recognition System](#)
Stack: MediaPipe, Scikit-learn, OpenCV, Python
 - Self-collected a dataset of **81,000 samples** across **26 ASL letters** using MediaPipe landmark extraction — storing structured keypoint data instead of raw images for efficiency.
 - Trained classical ML classifiers on MediaPipe hand landmarks, achieving **99% accuracy** with no signs of overfitting on the validation set.
 - Built a real-time recognition pipeline running at **30 FPS** via webcam, mapping live hand landmarks to predicted letters instantly.

- [AI Body Measurement System](#)

Stack: Computer Vision, Segmentation, PyTorch, Python, FastAPI

- Built a multi-output regression model predicting **14 body measurements** (chest, waist, hip, shoulders, etc.) from front and side view photos using segmentation masks and metadata.
- Achieved **val MAE of 2.9 cm** across all measurements — within acceptable range for clothing size estimation on real-world body images.
- Developed a web interface accepting gender, height, and weight as inputs alongside photos, returning measurements and clothing size recommendations instantly.

- [NYC Taxi Trip Duration Prediction](#)

Stack: Python, Scikit-learn, FastAPI

- Built a regression model predicting NYC taxi trip durations, achieving **R^2 of 0.63** using location and temporal features.
- Engineered domain-specific features including haversine distance, hour of day, and day of week to improve model performance.
- Deployed the model as a FastAPI REST API for real-time trip duration predictions.

[SKILLS](#)

- **ML & AI:** Supervised Learning, Unsupervised Learning, Deep Learning, Feature Engineering, Model Evaluation, Hyperparameter Tuning
- **Frameworks & Technologies:** PyTorch, FastAPI, OpenCV, YOLO, MediaPipe
- **Libraries:** NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn
- **Languages & Tools:** Python, C#, SQL, Docker, Git, GitHub, Linux
- **Languages:** Arabic, English

[COURSES & CERTIFICATIONS](#)

- **deeplearning.ai:** Machine Learning Specialization · Deep Learning Specialization · ML in Production
- **CSKilled:** Machine Learning Diploma